

Theory	BTH-64T-451 PLANT PHYSIOLOGY	CA	01 Hr	CA	20 Marks	CA	08 Marks
		EoSE	03 Hrs	EoSE	80 Marks	EoSE	32 Marks
Practical	BTH-64P-452 PLANT PHYSIOLOGY-PRACTICAL	CA	1 Hr	CA	10 Marks	CA	04 Marks
		EoSE	04 Hrs	EoSE	40 Marks	EoSE	16 Marks
Theory	BTH-64T-453 ANIMAL PHYSIOLOGY	CA	01 Hr	CA	20 Marks	CA	08 Marks
		EoSE	03 Hrs	EoSE	80 Marks	EoSE	32 Marks
Practical	BTH-64P-454 ANIMAL PHYSIOLOGY-PRACTICAL	CA	1 Hr	CA	10 Marks	CA	04 Marks
		EoSE	04 Hrs	EoSE	40 Marks	EoSE	16 Marks
Theory	BTH-64T-455 MOLECULAR GENETICS	CA	01 Hr	CA	20 Marks	CA	08 Marks
		EoSE	03 Hrs	EoSE	80 Marks	EoSE	32 Marks
Practical	BTH-64P-456 MOLECULAR GENETICS-PRACTICAL	CA	1 Hr	CA	10 Marks	CA	04 Marks
		EoSE	04 Hrs	EoSE	40 Marks	EoSE	16 Marks

The theory question paper will consist of **two** parts A&B.

PART-A: 20 Marks

Part A will be compulsory having 10 very short answer-type questions (with a limit of 20 words) of two marks each.

PART-B: 60 Marks

Part B of the question paper shall be divided into four units comprising question numbers 2-5. There will be one question from each unit with internal choice. Each question will carry 15 marks.

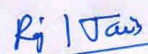
UG0804 – Three/Four Year Bachelor of Science (Bio-Technology)				
			Bio-Technology 2 nd YEAR SEM- IV	Credits

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#	L e v e l	S e m e s t e r	Type	Title	L	T	P	Total
1.	6	IV	MJR	UG0804 -BTH-64T-451 PLANT PHYSIOLOGY	4	0	0	4
2.	6	IV	MJR	UG0804 -BTH-64P-452 PLANT PHYSIOLOGY- PRACTICAL	0	0	2	2
3.	6	IV	MJR	UG0804 -BTH-64T-453 ANIMAL PHYSIOLOGY	4	0	0	4
4.	6	IV	MJR	UG0804 -BTH-64P-454 ANIMAL PHYSIOLOGY - PRACTICAL	0	0	2	2
5.	6	IV	MJR	UG0804 -BTH-64T-455 MOLECULAR GENETICS	4	0	0	4
6.	6	IV	MJR	UG0804 -BTH-64P-456 MOLECULAR GENETICS - PRACTICAL	0	0	2	2

BTH-64T-451 PLANT PHYSIOLOGY

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
IV	BTH-64T-451	PLANT PHYSIOLOGY			6	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Intermediate	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Basic Knowledge of Introductory/Foundation level				
Objectives of the Course		➤ This course aims to educate student about the mechanism and physiology life processes in plants. ➤ It focuses on the plant nutrient uptake and translocation, flowering and the roles of light in plant development.				


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Course Outcomes:

At the completion of the course, the student would be able to:

Cognitive level	Course outcomes
1. Understanding	➤ The various physiological life processes in plants. ➤ The various uptake and transport mechanisms in plants and are able to coordinate the various processes ➤ The role of various hormones, signaling compounds, thermodynamics and enzyme kinetics
2. Memorizing	➤ Principles and methods of various mechanisms in plants used for water transport ➤ How plants achieve water balance ➤ Assimilation of different mineral nutrients in plants ➤ The roles of light in plant development ➤ The plant hormones and their roles in plant development
3. Applying	➤ Find effective solutions for the treatment of mineral deficiency and problem associated.

Detailed Syllabus

BTH-64T-451 PLANT PHYSIOLOGY

UNIT-I

Plant water relationship: Water Potential and its components, water absorption by roots, aquaporins, pathway of water movement, symplast, apoplast, trans-membrane pathways, root pressure, guttation. Ascent of sap– cohesion-tension theory. Transpiration and factors affecting transpiration, anti-transpirants, mechanism of stomatal movement. Transpiration and its significance; Factors affecting transpiration; Root pressure and guttation.

Translocation in the phloem: Experimental evidence in support of phloem as the site of sugar translocation. Pressure–Flow Model; Phloem loading and unloading; Source– sink relationship.

15 Lectures

UNIT-II

Mineral nutrition: Essential and beneficial elements, macro and micronutrients, methods of study and use of nutrient solutions, criteria for essentiality, mineral deficiency symptoms, roles of essential elements, chelating agents.

Nutrient Uptake: Soil as a nutrient reservoir, transport of ions across cell membrane, passive absorption, electrochemical gradient, facilitated diffusion, active absorption, role of ATP, carrier systems, proton ATPase pump and ion flux, uniport, co-transport, symport, and antiport

15 Lectures

UNIT-III

Plant growth regulators: Discovery, chemical nature (basic structure), bioassay and physiological roles of Auxin, Gibberellins, Cytokinin, Absciscic acid, Ethylene. Brassinosteroids and Jasmonic acid.

Enzymes: Structure and properties; Mechanism of enzyme catalysis and enzyme inhibition. **15 Lectures**

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UNIT-IV

Physiology of flowering: Plant response to light and temperature: Photoperiodism (SDP, LDP, Day neutral plants); flowering stimulus, florigen concept, vernalization, seed dormancy.

Phytochrome: Discovery, chemical nature, role of phytochrome in photomorphogenesis, low energy responses (LER) and high irradiance responses (HIR), mode of action-red and far red light responses on photomorphogenesis.

15 Lectures

BTH-64P-452 PLANT PHYSIOLOGY-PRACTICAL

1. Determination of osmotic potential of plant cell sap by plasmolytic method.
2. Determination of water potential of given tissue (potato tuber) by weight method
3. To study the effect of two environmental factors (light and wind) on transpiration by excised twig.
4. Calculation of stomatal index and stomatal frequency
5. To study the phenomenon of seed germination (effect of light).
6. Demonstration of Hill reaction
7. To study the induction of amylase activity in germinating barley grains
8. Demonstrate the activity of catalase and study the effect of pH and enzyme concentration.
9. To study the effect of light intensity and bicarbonate concentration on O₂ evolution in photosynthesis
10. Comparison of the rate of respiration in any two parts of a plant.

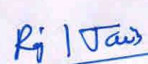
Suggested readings:

1. C. Sahu (2018). **Plant Physiology and Metabolism**. Kalyani Publishers, New Delhi.
2. Taiz, L., Zeiger, E., Møller, I.M. and Murphy, A (2015). **Plant Physiology and Development**. Sinauer Associates Inc. USA. 6th edition.
3. Hopkins, W.G., Huner, N.P., (2009). **Introduction to Plant Physiology**. John Wiley & Sons, U.S.A. 4th Edition.
4. Bajracharya, D., (1999). **Experiments in Plant Physiology- A Laboratory Manual**. Narosa Publishing House, New Delhi.
5. H. S. Srivastava. **Plant Physiology**, Rastogi Publications, New Delhi.

Course Learning Outcomes:

At the completion of the course, the student would be able to:

1. Understand plant water relationships, including water potential, absorption, movement, ascent of sap, transpiration, and factors affecting these processes.
2. Learn about translocation in the phloem, including experimental evidence, the pressure-flow model, phloem loading and unloading, and source-sink relationships.
3. Gain knowledge of mineral nutrition, essential and beneficial elements, criteria for essentiality, mineral deficiency symptoms, and roles of essential elements and chelating agents.
4. Comprehend nutrient uptake mechanisms, including ion transport across cell membranes, passive and active absorption, electrochemical gradients, and various carrier systems.
5. Explore plant growth regulators, their discovery, chemical nature, physiological roles, and the mechanisms of enzyme catalysis and inhibition.
6. Understand the physiology of flowering, plant responses to light and temperature, photoperiodism, the florigen concept, vernalization, seed dormancy, and the role of phytochrome


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in photomorphogenesis.

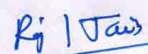
BTH-64T-453 ANIMAL PHYSIOLOGY

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
IV	BTH-64T-453	ANIMAL PHYSIOLOGY			6	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Intermediate	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Basic Knowledge of Introductory/Foundation level				
Objectives of the Course		➤ Students gain fundamental knowledge of animal physiology. ➤ Students are taught the detailed concepts of digestion, respiration, excretion, the functioning of nerves and muscles, cardiovascular system, endocrine system and reproductive system.				

Course Outcomes:

At the completion of the course, the student would be able to:

Cognitive level	Course outcomes
1. Understanding	➤ To understand the mechanisms that work to keep the animal body alive and functioning. ➤ To understand the physiological functions of various organ systems of the mammalian physiology. ➤ To understand the correlation between histology, anatomy and physiology
2. Memorizing	➤ Students learn the concepts of endocrine systems and homeostasis ➤ Comprehend the study of endocrine system their role in maintaining homeostasis of the human body ➤ The patho-physiology of common diseases related to organ systems of the body


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	The roles of light in plant development ➤
3. Applying	➤ Perform, analyze and report on experiments and observations in physiology ➤ The final goal of the course is to empower students to use their conceptual understanding to solve problems

Detailed Syllabus

BTH-64T-453 ANIMAL PHYSIOLOGY

UNIT-I

Introduction to Physiology: Cell & General Physiology, Membrane Physiology, Nerve and Muscle;
Blood and circulation: Blood corpuscles, haemopoiesis and formed elements, plasma function, blood volume, blood volume regulation, blood groups, haemoglobin, immunity, haemostasis, Immunity, and Blood Clotting. **The Heart and its Circulation:** Comparative anatomy of heart structure, myogenic heart, specialized tissue, ECG – its principle and significance, cardiac cycle, heart as a pump, blood pressure, neural and chemical regulation.

15 Lectures

UNIT-II

The Kidneys and Body Fluids: Comparative physiology of excretion, kidney, urine formation, urine concentration, waste elimination, micturition, regulation of water balance, blood volume, blood pressure, electrolyte balance, acid-base balance.

Gastrointestinal Physiology: Digestion, absorption, energy balance, BMR.

Respiration: Comparison of respiration in different species, anatomical considerations, transport of gases, exchange of gases, waste elimination, neural and chemical regulation of respiration, Aviation, Space, and Deep-Sea Diving Physiology.

15 Lectures

UNIT-III

The Nervous System: General Principles and Sensory Physiology, Neurons, action potential, gross neuroanatomy of the brain and spinal cord, central and peripheral nervous system, neural control of muscle tone and posture, Motor and Integrative Neurophysiology, the Special Senses.

Sense organs: Vision, hearing and tactile response.

15 Lectures

UNIT-IV

Endocrinology and Reproduction: Endocrine glands, basic mechanism of hormone action, hormones and diseases; reproductive processes and organs, neuroendocrine regulation.

Metabolism and Temperature Regulation: Comfort zone, body temperature – physical, chemical, neural regulation, acclimatization, Sports Physiology, Stress and adaptation.

15

Lectures

BTH-64P-454 ANIMAL PHYSIOLOGY-PRACTICAL



1. Preparation of blood film and microscopic study of stained blood film for identification of cell types.
2. Determination of RBC, WBC, PCV count in given blood sample.
3. Determination of haematocrit value of given blood sample.
4. Determination of haemoglobin content by Shali's haemometer.
5. Study of permanent slides of Mammalian tissue -To study TS of Thyroid gland, Ovary, kidney, bone, squamous epithelium, liver, blood smear, pancreas, muscle, small intestine, testis, nerve cell smear, trachea, VS of eye, stomach, skin, lung, brain, spinal cord.

Suggested readings:

1. **Textbook of Medical Physiology**, A C Guyton and John E Hall, W.B. Saunders Company, ISBN: 0-7216-5944-6.
2. **Fundamentals of Anatomy and Physiology**, FH Martini, Prentice Hall, ISBN: 0-13-017292-8
3. **Animal Physiology**, Prof. A. L. Bhatia & Dr. K. S. Kohli.
4. **Human Animal Physiology**, C. C. Chatterjee vol. 1&2.
5. **Animal Physiology**, Eckert.

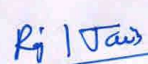
Course Learning Outcomes:

At the completion of the course, the student would be able to:

1. Understand cell and membrane physiology, nerve and muscle functions, blood components, circulation, and the mechanisms of immunity and blood clotting.
2. Comprehend the anatomy and physiology of the heart, including the cardiac cycle, ECG principles, heart function, blood pressure, and its neural and chemical regulation.
3. Learn the comparative physiology of excretion, kidney function, urine formation, and regulation of water, electrolyte, and acid-base balance.
4. Gain knowledge of gastrointestinal physiology, including digestion, absorption, energy balance, and respiration across species, with a focus on gas transport, exchange, and regulation.
5. Explore the nervous system, including the principles of sensory physiology, neuroanatomy, neural control of muscle tone, posture, and motor functions, and the special senses of vision, hearing, and tactile response.
6. Understand endocrinology and reproduction, the mechanisms of hormone action, reproductive processes, and neuroendocrine regulation, along with metabolism, temperature regulation, and physiological responses to stress and sports.

BTH-64T-455 MOLECULAR GENETICS

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
IV	BTH-64T-455	MOLECULAR GENETICS			6	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		


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Intermediate	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Basic Knowledge of Introductory/Foundation level				
Objectives of the Course		➤ Students gain the basics in genetic concepts and organization of genome on cellular and chromosomal level. ➤ Students are taught the basic molecular genetics mechanisms in relation to the structure and function of the cells.				

Course Outcomes:

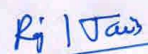
At the completion of the course, the student would be able to:

Cognitive level	Course outcomes
1. Understanding	➤ Acquire the knowledge in the field of Medical, Cyto-, Immuno-, Molecular, Cancer, Developmental and Neuro genetics to diagnose diseases and disorders. ➤ Capable to explain the various types of molecular biology methods that are used to study the regulation and function of biomolecules ➤ Acquire the ability to use their theoretical knowledge in solving practical issues. ➤ Know the bioethics and safety measures to be followed in handling the biological samples
2. Memorizing	➤ Understanding the fundamentals of hereditary materials and their role in functioning of human system. ➤ Able to identify the damage in hereditary material and malfunctioning of genes to help in eradicating the disease. ➤ Capable of understanding the Gene editing techniques ➤ Able to understand the human Genome and features ➤ With the wide technical knowledge, the students able to modify the genes and restore the functions of the hereditary material
3. Applying	➤ The final goal of the course is to empower students to use their conceptual understanding to solve problems

Detailed Syllabus

BTH-64T-455 MOLECULAR GENETICS

UNIT-I


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Central dogma and advancements: Properties and evolution of genetic material, flow of genetic information; Organization of viral, bacterial genomes and Eukaryotic genome; General concept of a gene, gene families, C-value paradox non- coding genes, repetitive DNA, Genome –types of genomes, genomes & genetic variation, comparison of different genomes, genome evolution. Genomics – about the genomics, history, comparative genomics, comparative genomic hybridization, functional genomics.

15 Lectures

UNIT-II

Fundamentals of DNA cloning and molecular hybridization: Cell based DNA cloning, vector-based cloning; nucleic acid hybridizations, PCR based DNA cloning and DNA analyses. Types of mutations and nomenclature, mutagenesis.

15 Lectures

UNIT-III

Recombination: Models and molecular mechanisms, Site Specific recombination: Molecular mechanism. Transposons and transposition mechanisms. Gene editing techniques: using CRISPR-Cas9, ZFNs and TALENs.

Features of the human genome: Organization of the human genome, human multigene families, Mapping of the human genome: Physical mapping and Genetic mapping. Footprints of evolution, human DNA instability. Chromosome walking. Introduction to human genome project- telomere to telomere, Ancestry by variations an overview of the project, goals of the project, major scientific strategies & approaches used in HGP, expected scientific & medical benefits of this project, about the organizations behind this project. Mapping of Human genome.

15 Lectures

UNIT-IV

Applications of molecular genetics: Disease diagnosis, Epigenetic testing, Prognostic and diagnostic markers, Development of molecules in Biopharma, Therapeutic advancements, Disease diagnosis and Disease inheritability, improving existing biological outcomes, Vaccine development and Gene therapy and other molecular genetics based therapeutic approaches.

15

Lectures

BTH-64P-456 MOLECULAR GENETICS-PRACTICAL

1. Introduction to Molecular genetics & Instrumentation.
2. Nucleic acid extraction: Isolation of prokaryotic DNA from bacterial cells.
3. Estimation of purity & quantity of extracted DNA by spectrophotometer.
4. Separation and estimation of integrity of DNA by agarose gel electrophoresis.
5. Determination of T_m values for the DNA sample.
6. Restriction Digestion and Ligation.
7. Primer designing.
8. Polymerase chain reaction.
9. Retrieval of sequences from nucleic acid databases.
10. Chromatogram analysis.

Suggested readings:

1. **Principles of Genetics**, Gardner, Simmons, Snustad 8th Edition 2006.



2. Tom Strachan and Andrew. P. Read, **Human Molecular Genetics**, Bios Scientific Pub UK (1996).
3. Watson, J.D., Hopkins, N.H., Roberts, J.W., Steitz, J. and Weinter, A.M., **Molecular Biology of Genes** (4th Edition) (1987). The Benjamin/Cummings publishing Company Inc., Joky.
4. Lewin, B. **Genes VI** (1997). Oxford University Press, Oxford, New York, Tokyo.
5. Darvell, J.et.al., **Molecular Cell Biology** (7th Edition) (2002). Garland Publishing Iwc., New York.
6. **Molecular Biology**, Glick and Pasternack (2003).
7. Lewin, **GenesIX**, 9th Edition Jones and Bartlett (2007).

Course Learning Outcomes:

At the completion of the course, the student would be able to:

1. Understand the central dogma of molecular biology, properties and evolution of genetic material, and the organization of various genomes.
2. Learn about genomics, including the history, comparative genomics, genomic hybridization, and functional genomics, as well as genome types, variations, and evolution.
3. Gain knowledge of DNA cloning techniques, as well as nucleic acid hybridizations and types of mutations.
4. Comprehend recombination mechanisms, site-specific recombination, transposons, and gene editing techniques.
5. Explore the organization and mapping of the human genome, the human genome project, and the scientific and medical benefits of genome research.
6. Understand applications of molecular genetics in various fields.

Raj Jais
 Dy. Registrar
 (Academic)
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